

STAT 516

Practice Midterm: Probability and Distributions

Name:_____ Please return this page with your solution after exam.

- 1.** Five cards are drawn at random from a 52-card deck.
 - (a). Compute the probability that at least two of them are spades.
 - (b). What is the probability that it will be a straight (Five cards in a sequence, regardless of suit. For example, 3,4,5,6,7 or 9,10, J, Q, K. Be careful that A can either be the minimum or the maximum).

- 2.** It is known that about 30% of human twins are identical and the rest are fraternal. Identical twins are necessarily the same sex — half are females and the other half are males. Furthermore, one-quarter of fraternal twins are both male, one-quarter are both female, and one-half are mixed (one male and one female).
 - (a). What is the probability that the randomly selected twins are twin boys?
 - (b). Given that a couple has just become the parents of twin girls, what is the probability the twins are identical?

- 3.** A continuous random variable X that has the probability density function

$$f(x) = \frac{1}{2}e^{-|x|}, \quad -\infty < x < \infty.$$

- (a). Find the median of X , i.e., find x such that $P(X < x) = \frac{1}{2}$ and $P(X \geq x) = \frac{1}{2}$.
- (b). Find the moment generating function of X and use it to determine the mean and standard deviation of X .
- (c). Let $Y = |X|$. Find the probability density of Y .

4. Suppose X is a continuous random variable with support $[1,5]$ and pdf

$$f(x) = \begin{cases} 0.5 - c|x - 3| & 1 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

(a). Find c .

(b). Find a such that $P(X \leq a) = 0.25$.

5. Consider the joint probability mass function

X_1	0	1	2
X_2			
1	0.2	0.1	0.1
2	0.2	0.0	0.1
3	0.0	0.0	0.3

(a). Compute $P(X_2 + X_1 > 3 | X_1 > 0)$.

(b). Compute $E(X_1), E(X_2), Var(X_1)$ and $Var(X_2)$.

(c). Compute the correlation coefficient between X_1 and X_2 .

(d). Compute $E[\max(X_1, X_2)]$.

6. Suppose X and Y are independent *Exponential* random variables with parameter λ , i.e., they both have the density

$$f(x) = \frac{1}{\lambda} e^{-x/\lambda}, \quad x > 0, \quad \lambda > 0.$$

Let $U = X + Y$ and $V = X/Y$.

(a). Find the joint density of U and V .

(b). Are U and V independent? Why?

(c). Find the marginal density of U and V .

(d). Find $E(V|U = 1)$.

Bonus. A real number is picked at random from $[0, 1]$. Verify that event A that the first decimal digit of the number picked is 5 is independent of event B that the third decimal digit is 1.